

Curcumin (diferuloylmethane) down-regulates the constitutive activation of nuclear factor-kappa B and IkappaBalpha kinase in human multiple myeloma...

Because of the central role of the transcription factor nuclear factor-kappaB (NF-kappaB) in cell survival and proliferation in human multiple myeloma (MM), we explored the possibility of using it as a target for MM treatment by using curcumin (diferuloylmethane), an agent known to have very little or no toxicity in humans. We found that NF-kappaB was constitutively active in all human MM cell lines examined and that curcumin, a chemopreventive agent, down-regulated NF-kappaB in all cell lines as indicated by electrophoretic mobility gel shift assay and prevented the nuclear retention of p65 as shown by immunocytochemistry. All MM cell lines showed constitutively active IkappaB kinase (IKK) and IkappaBalpha phosphorylation. Curcumin suppressed the constitutive IkappaBalpha phosphorylation through the inhibition of IKK activity. Curcumin also down-regulated the expression of NF-kappaB-regulated gene products, including IkappaBalpha, Bcl-2, Bcl-x(L), cyclin D1, and interleukin-6. This led to the suppression of proliferation and arrest of cells at the G(1)/S phase of the cell cycle. Suppression of NF-kappaB complex by IKKgamma/NF-kappaB essential modulator-binding domain peptide also suppressed the proliferation of MM cells. Curcumin also activated caspase-7 and caspase-9 and induced polyadenosine-5'-diphosphate-ribose polymerase (PARP) cleavage. Curcumin-induced down-regulation of NF-kappaB, a factor that has been implicated in chemoresistance, also induced chemosensitivity to vincristine and melphalan. Overall, our results indicate that curcumin down-regulates NF-kappaB in human MM cells, leading to the suppression of proliferation and induction of apoptosis, thus providing the molecular basis for the treatment of MM patients with this pharmacologically safe agent.